Project Architecture & System Specification

Title

Resolving Parametric Knowledge Conflicts via Temporal Metric Learning and Residual LLM Probing

1. Problem Statement & Motivation

When facts mutate over time (e.g., corporate policies, regulatory guidelines, software dependencies), Large Language Models (LLMs) suffer from two distinct failure modes:

1. **Retrieval Degradation (Keyword Overlap):** Standard dense vector retrievers map temporally stale documents close to user queries because they share high lexical and semantic overlap with fresh documents.
2. **Parametric Prior Dominance (LLM Hallucination):** Even when supplied with updated contextual passages via RAG, the LLM's internal pre-trained memory often overrides the retrieved context during autoregressive decoding, generating outdated responses.

This project addresses both failure modes without requiring full parameter fine-tuning of the generator model.

2. System Architecture Overview

```
                     [ User Query (q) ]
                             │
                             ▼
      ┌───────────────────────────────────────────┐
      │   STAGE 1: Time-Aware Metric Learning       │
      │   - Custom Bi-Encoder (MiniLM / BGE-Small)  │
      │   - Margin-Penalized InfoNCE Loss             │
      └─────────────────────┬─────────────────────┘
                             │
                             ▼
                [ Top-1 Fresh Passage (d+) ]
                             │
                             ▼
      ┌───────────────────────────────────────────┐
      │   STAGE 2: Prompt Construction & Forward    │
      │   - System Prompt + Context (d+) + Query (q) │
      │   - Base Generator (Qwen-2.5-1.5B / Llama-3.2)│
      └─────────────────────┬─────────────────────┘
                             │
                  [ Residual Stream Extraction ]
                             │
                             ▼
      ┌───────────────────────────────────────────┐
      │   STAGE 3: Mechanistic Conflict Probing     │
      │   - Extract Hidden States h_l at Layer l   │
      │   - Linear Probe: P(Conflict) = σ(W^T h_l)  │
```

```
                   └────────────────────────┬─────────────────────────┘
                                            │
                              Is P(Conflict) > τ ?
                                  ┌─────────┴──────────┐
                              YES │                    │ NO
                                  ▼                    ▼
            ┌──────────────────────┐    ┌──────────────────────┐
            │ STAGE 4: Steered       │    │ Standard Autoregressive│
            │ Contrastive Decoding   │    │ Greedy Generation      │
            │ z_steered = z_ctx -    │    │ z_final = z_base       │
            │           α * z_param  │    └──────────┬───────────┘
            └──────────┬───────────┘               │
                       │                              │
                       └─────────────┬─────────────┘
                                     │
                                     ▼
                            [ Verified Response ]
```

3. Deep-Dive Component Specifications

Component 1: Time-Aware Dual-Encoder (Metric Learning)

Objective

Train a bi-encoder $(\mathbf{E}_Q, \mathbf{E}_D)$ to project queries and documents into a shared latent space $\mathbb{R}^d$ where fresh documents are separated from temporally stale documents, even when lexical overlap is near 1.0.

Loss Formulation

We modify the InfoNCE contrastive loss by adding an explicit distance margin $\gamma > 0$ for temporally invalid hard negatives:

$$\mathcal{L}_{\text{temporal\_NCE}} = -\log \frac{\exp\left(\frac{\mathbf{e}_q \cdot \mathbf{e}_{d^+}}{\tau}\right)}{\exp\left(\frac{\mathbf{e}_q \cdot \mathbf{e}_{d^+}}{\tau}\right) + \sum_{j=1}^{N} \exp\left(\frac{\mathbf{e}_q \cdot \mathbf{e}_{d_j^-}}{\tau}\right) + \sum_{k=1}^{K} \exp\left(\frac{\mathbf{e}_q \cdot \mathbf{e}_{d_k^c} + \gamma}{\tau}\right)}$$

Where:

- $\mathbf{e}_q = \frac{\mathbf{E}_Q(q)}{\|\mathbf{E}_Q(q)\|_2}$ (Normalized query embedding)
- $\mathbf{e}_{d^+} = \frac{\mathbf{E}_D(d^+)}{\|\mathbf{E}_D(d^+)\|_2}$ (Normalized fresh document embedding)
- $\mathbf{e}_{d_j^-}$ are random in-batch negative embeddings.
- $\mathbf{e}_{d_k^c}$ are mined **temporally stale/conflicting** hard negatives.
- $\gamma = 0.2$ is the enforced margin penalty.
- $\tau = 0.05$ is a learned temperature scaling factor.

Latent Geometry Metrics

To verify the latent space resists dimensional collapse, we calculate:

- **Alignment:**

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(q,d^+)} \left[\|\mathbf{e}_q - \mathbf{e}_{d^+}\|^2 \right]$$

- **Uniformity:**

$$\mathcal{L}_{\text{uniform}} = \log \mathbb{E}_{x,y \sim p_{\text{data}}} \left[\exp\left(-2\|\mathbf{e}_x - \mathbf{e}_y\|^2\right) \right]$$

Component 2: Mechanistic Residual Stream Probing

Objective

Detect when the generator LLM undergoes internal cognitive conflict between its pre-trained parametric memory and the retrieved context during token generation.

Layer Extraction Methodology

1. Pass the prompt $P = (d^+, q)$ into the autoregressive LLM.
2. Hook into intermediate transformer layers $l \in [1, L]$ to extract hidden activations $\mathbf{h}_l^{(t)} \in \mathbb{R}^{d_{\text{model}}}$ at generation step t.
3. Train a lightweight logistic regression probe $\mathbf{W}_l \in \mathbb{R}^{d_{\text{model}}}$ per layer:

$$\hat{y}_{\text{conflict}}^{(t)} = \sigma\left(\mathbf{W}_l^T \mathbf{h}_l^{(t)} + b_l\right)$$

```
Layer L-4:  [ Token Embeddings ] ──► [ Attention / FFN ] ──► h_{L-4} (Low Seman
Layer L/2:  [ Residual Stream ] ──► [ Linear Probe ]   ──► P(Conflict) ≈ 0.85 (H
Layer L:    [ Logit Projection] ──► [ Unsteered Output ]──► Parametric Hallucin
```

Ground Truth Synthetic Probing Dataset

- **Class 0 (No Conflict):** Query matches both parametric memory and context.
- **Class 1 (Parametric Conflict):** Context explicitly contradicts high-confidence parametric pre-training facts (e.g., mutated dates, updated policy numbers).

Component 3: Dynamic Contrastive Logit Decoding (Inference Steering)

Objective

When $\hat{y}_{\text{conflict}}^{(t)} > \tau_{\text{threshold}}$, dynamically steer output logits to prioritize contextual information over memorized priors without updating base weights.

Mathematical Formulation

Compute two forward passes during generation:

1. **Contextual Forward Pass:** Input contains full context + query $\rightarrow$ Logits $z_{\text{context}}^{(t)}$.
2. **Null/Parametric Forward Pass:** Input contains query only (no context) $\rightarrow$ Logits $z_{\text{parametric}}^{(t)}$.

Formulate steered probability distribution:

$$p_{\text{steered}}(y_t) = \text{softmax}\left(z_{\text{context}}^{(t)} - \alpha \cdot z_{\text{parametric}}^{(t)}\right)$$

Where $\alpha \in [0.1, 0.5]$ is the dynamic steering coefficient adjusted according to probe confidence:

$$\alpha = \alpha_0 \cdot \hat{y}_{\text{conflict}}^{(t)}$$

4. Operational Step-by-Step Execution Lifecycle

Step	**Phase**	**Input**	**Action**	**Output**
1	**Ingestion**	User Query q, Document Corpus $\mathcal{D}$	Encode corpus and query via dual-encoder fine-tuned with $\mathcal{L}_{\text{temporal\_NCE}}$.	Top-1 passage d^+
2	**Prompting**	q, d^+	Construct prompt: `"Context: {d+} \n Query: {q} \n Answer:"`	Input tokens $X_{1:N}$
3	**Forward Pass**	Tokens $X_{1:N}$	Compute base transformer forward pass up to layer l^*.	Hidden state $\mathbf{h}_{l^*}^{(N)}$
4	**Probe Check**	$\mathbf{h}_{l^*}^{(N)}$	Evaluate trained linear classifier $\sigma(\mathbf{W}_{l^*}^T \mathbf{h}_{l^*}^{(N)} + b)$.	Conflict Probability P_c
5	**Branching**	P_c	If $P_c > 0.50$, trigger double forward pass for logit contrast. Else, use baseline greedy decoding.	Output Logits $z^{(t)}$
6	**Decoding**	$z^{(t)}$	Sample token $y_t \sim p(y_t)$ and append to sequence. Repeat until `EOS`.	Final String Response

5. Evaluation & Science Metrics

System Sub-Module	**Scientific Metric**	**Target Baseline**	**Proposed Method Target**
Retrieval Engine	Mean Reciprocal Rank (MRR@1)	0.62 (Contriever)	**> 0.85** ($\mathcal{L}_{\text{temporal\_NCE}}$)
Retrieval Engine	Representation Uniformity	-1.20 (Standard BERT)	**< -2.20** (High Isotropy)
Probing Module	ROC-AUC (Conflict Detection)	0.50 (Random Chance)	**> 0.88** (Layer 18–24 Probes)
Generative Pipeline	Context Reliance Rate (CRR)	58.2% (Base Qwen/Llama)	**> 88.0%** (Contrastive Steering)
Generative Pipeline	Hallucination Rate	34.0%	**< 10.0%**

6. Lightweight Hardware & Resource Setup

- **Retriever Model:** `sentence-transformers/all-MiniLM-L6-v2` (22M parameters)
- **Generator Model:** `Qwen/Qwen2.5-1.5B-Instruct` or `meta-llama/Llama-3.2-1B-Instruct`

- **GPU Memory Footprint:**
  - Dual-encoder fine-tuning (`fp16` , batch size 32, grad accumulation 4): **~3.2 GB VRAM**
  - LLM inference & hidden state probing (`4-bit NF4` or `fp16` batch size 1): **~4.5 GB VRAM**
- **Total Execution Cost:** $0 (Executable on free Kaggle / Google Colab T4 GPUs).